

Research and Engineering Report

Biomechanical Feature Extraction and Deep Learning Architectures for Gait Anomaly Localization

1. Executive Summary

This report presents a deep learning framework designed to transition raw 3D pose data into clinical biomechanical insights for gait analysis and anomaly localization. The pipeline captures continuous, time-series human movement and processes it through two distinct unsupervised deep learning methodologies: an LSTM Autoencoder and an AnoGAN-style Generative Adversarial Network (GAN).

Both models are trained entirely on healthy, typical gait signatures to learn a robust baseline representation of normal human locomotion. Anomalies are subsequently detected and localized by measuring structural and temporal deviations (reconstruction and adversarial residual errors) when the networks are exposed to unseen, pathological data.

2. Data Pipeline & Biomechanical Feature Engineering

Before neural processing, raw geometric coordinate inputs are transformed into structured, medically relevant biomechanical time-series features.

2.1 Preprocessing & Windowing

- **Standardization:** Raw 3D skeletal coordinates (x, y, z) tracking 8 primary anatomical joints (shoulders, hips, knees, and ankles) are scaled utilizing a StandardScaler fitted exclusively on the normal walking profiles to stabilize variance distributions.
- **Sliding Window Segmentation:** To capture the continuous, cyclic patterns of human locomotion, the data is sliced into sliding temporal windows with a fixed sequence length of $T = 50$ frames. An off-by-one boundary protection strategy isolates individual data files during segmentation to prevent temporal leakage between distinct recordings.

2.2 Feature Augmentation Matrix

The initial 24 spatial coordinates (8 joints $\times$ 3 dimensions) are expanded into a 48-dimensional feature vector per frame by extracting five categories of engineered biomechanical properties:

- **Shoulder-Hip Coupling ($X_{\text{shoulder}} - X_{\text{hip}}$):** Evaluates rotational coordination and rigid alignment between the shoulder and pelvic girdles.
- **Left-Right Symmetry ($|X_{\text{left_joint}} - X_{\text{right_joint}}|$):** Highlights limping, favored sides, or lateralized neuromotor impairments.
- **Contralateral Coordination (Cross-difference vectors):** Captures the reciprocal, cross-body swinging mechanics fundamental to natural human locomotion.

- **Torso Sway ($1/2 * (X_{\text{left_shoulder}} + X_{\text{right_shoulder}})$):** Tracks upper-body instability, shifts in equilibrium, or center-of-mass balancing compensations.
- **Trunk Inclination (Hip center to torso sway center):** Quantifies forward/backward leaning postures (e.g., stooped gait profiles).

3. Deep Learning Architectures: Mechanisms & Operations

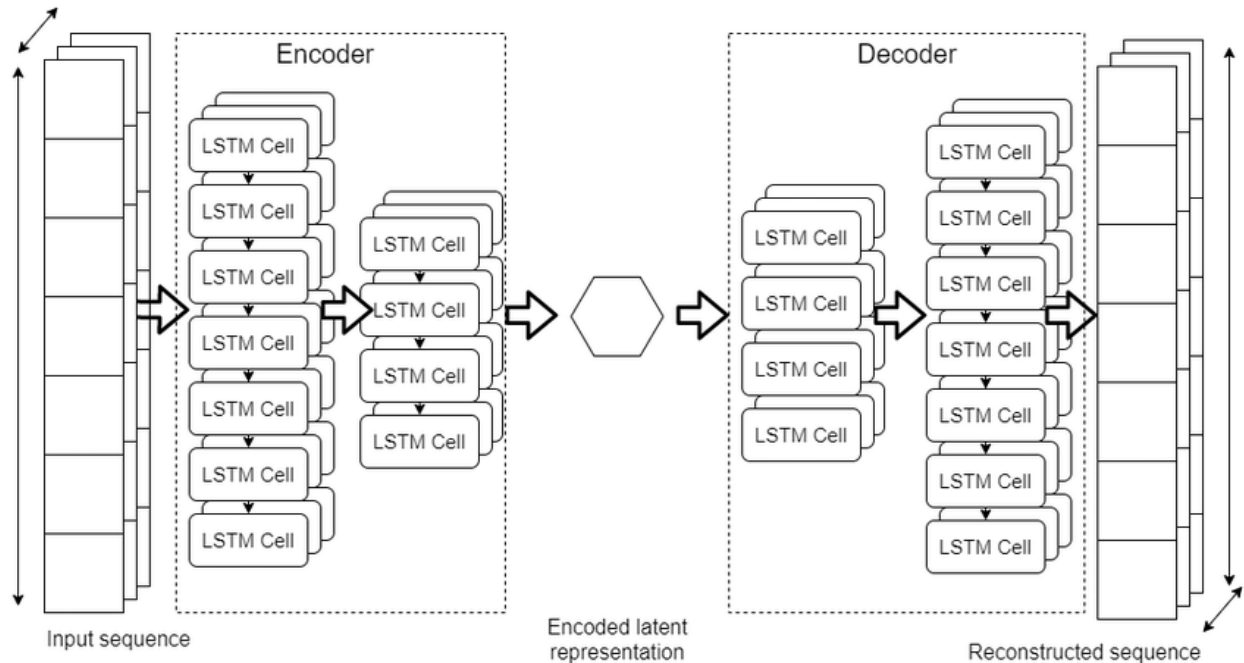
The framework evaluates two alternative neural network approaches to process the 48-dimensional temporal windows. While both serve the purpose of unsupervised anomaly detection, their internal data handling and processing mechanisms differ.

3.1 Architecture A: LSTM Autoencoder

The LSTM Autoencoder utilizes sequential compression and decompression to model temporal dependencies via a dedicated Encoder-Decoder workflow:

- **The Encoder:** A Long Short-Term Memory (LSTM) layer ingests the sequential data step-by-step. At each frame, the cell updates its internal hidden state and cell memory state, which selectively retain historical information across time steps.
- **Latent Bottleneck:** The final outputs of the encoder's sequence, representing the aggregated temporal context of the complete gait cycle, serve as a compressed information bottleneck.
- **The Decoder:** Another LSTM layer unrolls the bottleneck representation back into a 50-frame window. To prevent the network from simply copying identity sequences, the decoder operates autoregressively: it receives the final frame of the input sequence as its initial token, outputs an estimated frame via a Linear transformation layer, and feeds that prediction forward as the input for the next generation step.
- **Temporal Alignment:** The output list is inverted along the time dimension via `torch.flip` to structurally realign the output sequence chronology with the target window.

Objective Function: Optimized using Mean Squared Error (MSE) Loss across the sequence windows to ensure normal profiles are meticulously learned.



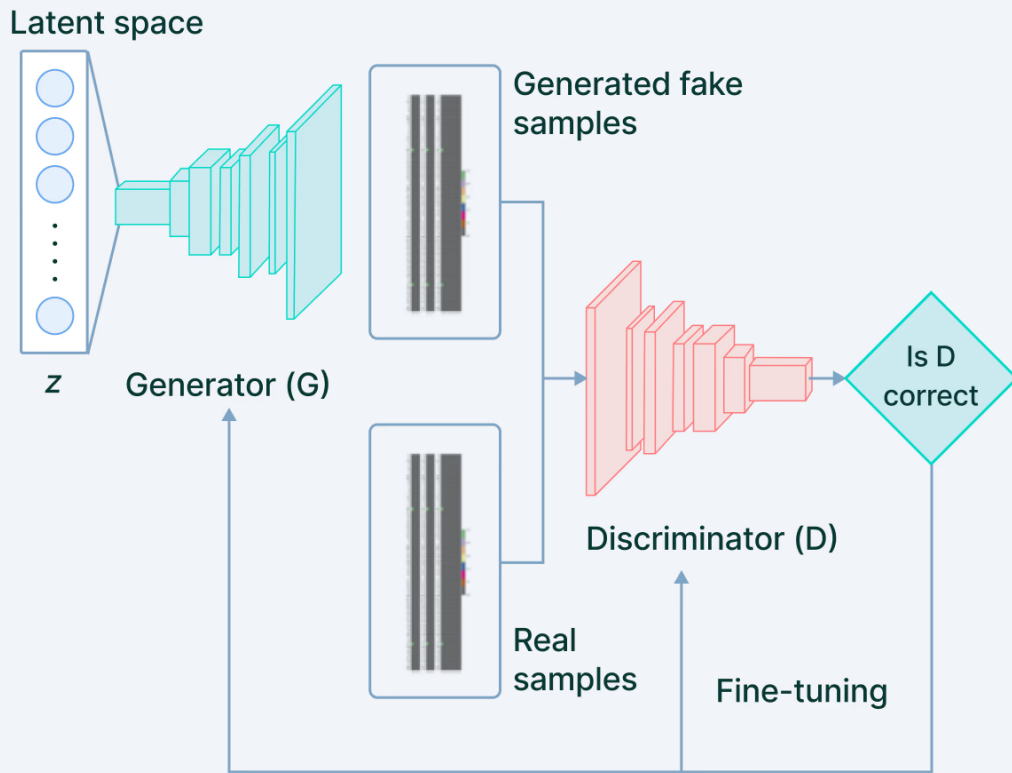
3.2 Architecture B: Generative Adversarial Network (GAN)

The adversarial pipeline structures anomaly detection as a game between two competing networks: a Generator (G) and a Discriminator (D).

- **The Generator (G):** Ingests a randomized, low-dimensional latent noise vector z in $\mathbb{R}^{64}$. This vector passes through a fully connected network (`nn.Sequential`) that expands the tensor's dimensions, reshapes it into a sequential framework (50×32), and routes it through a two-layer LSTM network. A concluding Linear layer maps the hidden states into a fully synthetic 50×48 gait matrix.
- **The Discriminator (D):** Acts as an evaluation network. It accepts a gait window (either real from the dataset or fake from G) and processes it through an LSTM cell. The final hidden state passes through a classifier sequence with LeakyReLU activations to output a scalar validity score indicating whether the sequence matches the true distribution.

Optimization Protocol: Implemented via `nn.BCEWithLogitsLoss` where the Discriminator maximizes true training window validation scores, while the Generator learns to output realistic coordinates to counter the Discriminator's screening rules.

Generative Adversarial Network



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4. Architecture Comparison & Paradigm Differentiation

While both models parse the same data, they rely on entirely different mathematical frameworks:

Architectural Attribute	LSTM Autoencoder (AE)	Generative Adversarial Network (GAN)
Primary Learning Objective	Deterministic Reconstruction: Learns to compress and accurately recreate input features.	Distribution Mapping: Learns to map random noise configurations onto realistic feature trajectories.
Model Composition	Encoder LSTM, Decoder LSTM, and	Generator (Linear + LSTM) and

	a Linear output layer.	Discriminator (LSTM + Classifier).
Latent Space Interpretation	Continual embedding space derived directly from actual physical input windows.	Stochastic latent space representing an underlying motion distribution.
Information Bottleneck	Explicit compression bottleneck (128 hidden nodes) forcing feature reduction.	Implicit; defined by the adversarial feedback loop between G and D.
Sensitivity Profile	Highly sensitive to point-to-point coordinate shifts and precise localized trajectory deviations.	Highly sensitive to global stylistic consistency, structural coherence, and natural rhythms.

5. Benefits of the Strategy: Training on Normal Data Only

Choosing to train recurrent architectures (LSTMs) exclusively on a baseline of normal data yields major strategic advantages for biomechanical engineering and clinical deployment:

5.1 Elimination of the 'Anomalous Data Scarcity' Bottleneck

In clinical environments, healthy walking patterns are abundant and easily recorded, whereas pathological gaits (resulting from rare neurological conditions, specific injuries, or unique post-operative recovery paths) are highly scarce, diverse, and difficult to collect. By framing this as a one-class anomaly detection problem, the system circumvents the need for massive, balanced datasets of injuries.

5.2 Zero-Shot Detection of Unseen Pathologies

A supervised classifier trained to identify specific gait abnormalities (e.g., a stroke-induced hemiparetic gait) would fail or misclassify an entirely new type of pathology (e.g., an acute ankle sprain or Parkinsonian tremors). Because the Autoencoder and GAN only understand normal locomotion, they recognize any deviation from healthy mechanics as an anomaly, allowing the system to flag completely novel or uncatalogued walking impairments.

5.3 High Temporal Fidelity with LSTMs

Gait anomalies are rarely constant; they manifest at specific instances in time, such as during the heel-strike or toe-off phase. Traditional fully connected or static networks miss these temporal changes. By choosing LSTM layers, the models map the explicit velocity, acceleration, and sequential harmony of a normal step. This makes it easy to isolate exactly when the rhythm is broken.

6. Anomaly Detection & Localization Framework

Because both networks are trained exclusively on healthy gait profiles, they lack baseline representations for pathomechanics or limping styles. Anomaly localization exploits this limitation.

6.1 Autoencoder Anomaly Scoring

When the Autoencoder evaluates an abnormal sequence, it attempts to reconstruct the data using patterns learned from normal gaits. As a result, anomalous movements are reconstructed with high error rates. This difference creates an isolated error profile tracking pointwise variations.

6.2 GAN (AnoGAN-Style) Mapping Optimization

Because a GAN does not possess an explicit encoding layer to ingest input data, an iterative latent space optimization procedure is required during inference:

- An input sequence X is isolated, and a mutable latent tensor z is initialized.
- z passes through the Generator to yield a synthetic window, $G(z)$.
- A dual-component loss function is calculated balancing Residual Loss (absolute coordinate differences) and Discrimination Feature Matching Loss (structural hidden-state anomalies via D 's layers).
- Backpropagation optimizes the vector z rather than the model weights, iteratively adjusting it over several hundred steps to find the closest possible match within the learned normal distribution.

6.3 Diagnostic Localization Engine

Once error profiles are established by either model, a diagnostic engine converts them into clear physiological insights:

- **Temporal Tracking:** Rows are aggregated across all 48 features to track exactly when an anomaly occurs within the 50-frame window.
- **Biomechanical Profiling:** Columns are grouped according to their specific indices using the project's anatomical feature mapping dictionary (JOINT_MAP). By calculating mean errors across specialized feature groups, the framework pinpoints exactly which joints or biomechanical variables (e.g., left ankle deviation versus right-hip symmetry) are causing the abnormal gait profile.